

BITTUXEROXKIOSK

15,000+
students. 1 photocopy shop.
Closes at 5:30 PM.

Autonomous 24/7 Smart Self-Service Printing & Document Kiosk

Zero Queues · Zero Leakage · 24/7 Always On

THE CRISIS

Your Aadhaar on a stranger's phone.



9:30 AM – 5:30 PM

Restricted Operating Hours



20–45 Minutes

Peak Submission Queue Time



₹3 – ₹5 / Page

Monopoly Manual Pricing

THE SOLUTION

Meet Bittu.



24/7

Always On

₹2.50

Per Page

0.36 m²

Footprint



Secure

Zero Data Retention

UNDER THE HOOD

Engineering That No Competitor Has.

⚡ EXPRESS LANE SCHEDULER

Small jobs never wait.

- Bulk jobs split into 5–page micro–chunks.
- Between chunks, express jobs (< 5 pages) jump in.

100–page job running. 2–page job arrives:

[Chunk 1-5] → PAUSE → [2-pg Express] → RESUME [6-10]...

Student B waits 15 sec, not 8 minutes.

🔋 3-TIER SMART POWER

71% energy savings via PIR.

3 min idle

Display dims, printer standby

22W

15 min idle

Display OFF, deep sleep

4W

2–6 AM

Night mode, relay cuts power

1.5W

🔒 DPDP-COMPLIANT SECURITY

Volatile RAM Buffer (tmpfs)

Files live in RAM only — never touch disk.

Post-Print Auto-Delete

Removed the instant CUPS says done.

TLS 1.3 In-Transit

End-to-end encrypted file transfer.

🏫 CAMPUS ERP INTEGRATION

1–Tap Exam Hall Tickets

Print admit cards with security watermark.

Grade Cards & Certificates

Semester reports, fee receipts, attendance.

Secure Session Bridge

Student authenticates via Reg No + OTP.

🔧 TECH STACK

Frontend: **Chromium Kiosk PWA**

Backend: **Python + FastAPI**

Engine: **CUPS on Linux**

Bot: **python-telegram-bot**

Hardware: **Raspberry Pi 5 / 4B**

WORKFLOW

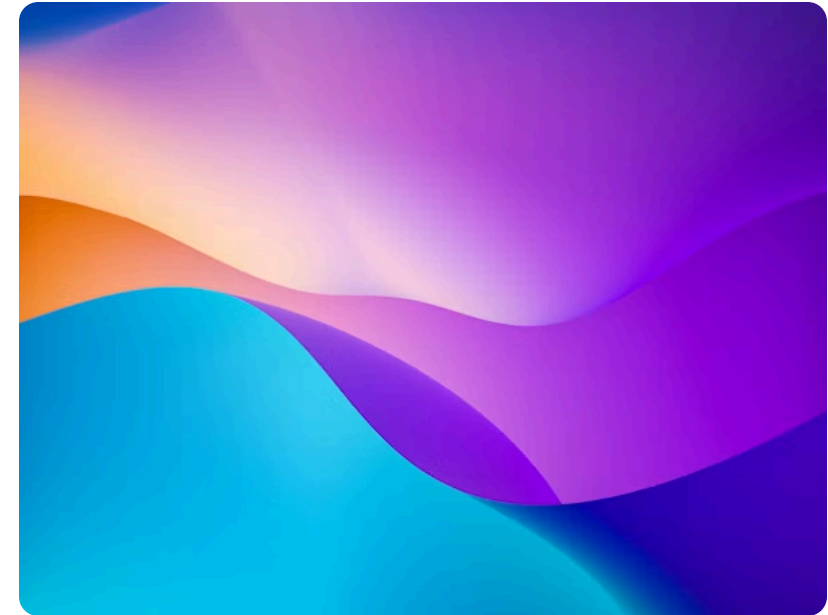
Scan. Upload. Pay. Print.

Under 60 seconds. No app download. No clerk. No queue.



QR + Web Upload

No app install needed



Telegram / WhatsApp Bot

Send doc → get PIN → print

The Numbers Work.

COST BREAKDOWN PER PAGE

Paper (75 GSM)	₹0.52 (43%)
Toner Cartridge	₹0.25 (21%)
Drum Amortization	₹0.10 (8%)
Electricity	₹0.03 (3%)
Jam/Misprint Buffer	₹0.05 (4%)
Campus Floor Share	₹0.25 (21%)

Total Production Cost ₹1.20

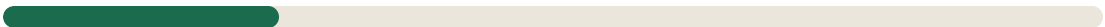
COST → SELL = PROFIT
₹1.20 → ₹2.50 = ₹1.30

52% Gross Margin

CAPEX: ₹48,400

MONTHLY PROFIT & PAYBACK

Conservative · 150 pages/day ₹4,875/mo



Payback: ~10.5 months

Realistic · 350 pages/day ₹11,375/mo



Payback: ~4.2 months

Peak Season · 700 pages/day ₹22,750/mo



Payback: ~2.1 months ⚡

COMPETITIVEEDGE

70% Lower. 100% Yours.

Qwikprint PRO

₹3.49 Lakh

Proprietary · Revenue Cut · No Campus API

Qwikprint MINI

₹1.65 Lakh

Limited features

★ Bittu Xerox Kiosk

₹50,000

100% Owned



100% Profit

No royalty splits



Open Source

Python + FastAPI



Campus-Native

ERP + Telegram Bot

FULL 3D CAD · ENGINEERED & MODELED

Parametric OpenSCAD 3D Assembly + 2D AutoCAD Sheet Metal Blueprints

T H E A S K

Let's Build Together.

₹1 Lakh

Seed Grant
for 2 Pilot Kiosks

0.36 m²

Lobby Space
+ 1 Power Socket

~30 Days

To Deliver
Working Pilot

Start with 1 kiosk at KLU. Scale to **~50 campuses** across AP & Telangana.

Every exam season without a kiosk = 2,00,000+ pages of unserved demand.

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